

MANSI SRINIVAS

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SUMMARY

- Software Engineer with **3+ years of experience** designing and delivering **Python-based backend systems** and **microservices architectures** for enterprise-scale applications across insurance and technology domains.
- Strong expertise in **Python (FastAPI, Flask, Django)** and **RESTful / GraphQL APIs**, building secure, high-performance services handling **millions of transactions and API requests** in cloud-native environments.
- Hands-on experience with **AWS cloud services** (Lambda, ECS, EKS, API Gateway, S3) and **containerized deployments** using **Docker and Kubernetes**, enabling scalable, fault-tolerant production systems.
- Proven ability to design and optimize **relational and NoSQL databases** including **PostgreSQL, DynamoDB, MySQL, and MongoDB**, improving query performance, data integrity, and system reliability.
- Experienced in implementing **CI/CD pipelines**, automated testing, and **DevOps best practices** using **GitHub Actions, Jenkins, PyTest, and Selenium**, reducing deployment cycles and production defects.
- Effective collaborator within **Agile/Scrum environments**, working closely with product owners, QA, and cross-functional teams to deliver high-quality solutions aligned with business and compliance requirements.

SKILLS

Programming Languages: Python, SQL, JavaScript, TypeScript, Java, Bash

Front-End Technologies: React.js (18+), Hooks, Context API, HTML5, CSS3, Tailwind CSS, Bootstrap, Material-UI, AJAX, Responsive UI Development

Back-End Technologies: Python (FastAPI, Flask, Django), RESTful APIs, Microservices Architecture, GraphQL APIs, gRPC, WebSockets, Asynchronous Programming, OAuth2

Cloud & DevOps: AWS (Lambda, ECS, EKS, EC2, S3, RDS, DynamoDB, API Gateway), Azure (Functions, AKS), Docker, Kubernetes, Terraform, Jenkins, GitHub Actions, CI/CD Pipelines

Databases & Messaging: PostgreSQL, MySQL, SQL Server, MongoDB, DynamoDB, Redis, Cassandra, Apache Kafka, **Testing &**

Quality Engineering: PyTest, Unit Testing, Integration Testing, Selenium, Postman, Swagger/OpenAPI

Monitoring & Observability: Prometheus, Grafana, ELK Stack, OpenTelemetry, Splunk, AppDynamics

Development Tools: Git, GitHub, GitLab, Maven, Gradle, Jira, Confluence, Nginx

EXPERIENCE

IBM, USA

Jun 2025 – Present

Software Engineer

- Design and develop backend microservices using **Python (FastAPI, Flask)** within a **microservices architecture**, supporting **10M+ monthly API calls** and improving average response times by **35%** for customer-facing insurance platforms.
- Build cloud-native services on **AWS (Lambda, ECS, EKS, API Gateway, S3)** using **Docker and Kubernetes**, enabling horizontal scalability and reducing infrastructure provisioning and deployment effort by **40%**.
- Implement secure **RESTful and GraphQL APIs** with **OAuth2**, strengthening authentication and authorization flows while reducing security vulnerabilities and access-related defects by **30%**.
- Design and optimize data persistence layers using **PostgreSQL and DynamoDB**, improving query execution time and transactional throughput by **25%** under high-concurrency workloads.
- Establish automated **CI/CD pipelines** using **GitHub Actions and Jenkins**, integrating unit and integration tests to reduce release cycles from days to **under one hour** with minimal manual intervention.
- Implement application monitoring and observability using **Prometheus, Grafana, and OpenTelemetry**, improving system visibility and reducing mean time to resolution (MTTR) for production issues by **45%**.

Cisco, India

Jan 2021 - Jul 2023

Software Engineer

- Developed backend services using **Python (Django, Flask)** and **RESTful APIs**, supporting business applications used by **100K+ users** and improving API stability and performance by **30%**.
- Built responsive front-end components using **React.js, JavaScript (ES6+), HTML5, and CSS3**, enhancing user experience and reducing page load times by **25%** across client-facing modules.
- Designed and maintained relational and NoSQL databases using **MySQL, PostgreSQL, and MongoDB**, improving data consistency and reducing application-level data errors by **20%**.
- Implemented asynchronous processing and messaging using **Apache Kafka and RabbitMQ**, enabling event-driven communication and reducing background job processing latency by **35%**.
- Developed automated test suites using **PyTest, and Selenium**, increasing test coverage to **80%** and reducing post-release production defects by **25%**.
- Collaborated with product managers and QA teams using **Jira and Confluence** within **Agile frameworks**, ensuring timely delivery of features and improving sprint predictability by **20%**.

EDUCATION

Master of Science in Computer Science | State University of New York at Binghamton

December 2025

Bachelor of Computer Science | Visvesvaraya Technological University (VTU)

Jul 2023